# Faizan Bari

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## Education

2025–present **Institute of Mathematical Sciences (IMSc), Chennai**
Integrated PhD programme in theoretical physics, Master's stage.
First-year courses (grades out of 10): Quantum Mechanics I (10), Quantum Mechanics II (9), Classical Mechanics (9), Electrodynamics (9), Statistical Mechanics I (9), Condensed Matter Physics (9), Computational Mechanics (9), Mathematical Physics (8).
In progress: Black Holes, Holography and Quantum Information; Mathematical Methods II.

2024 **National Institute of Technology (NIT), Srinagar**
B.Tech in Electronics and Communication Engineering. CGPA 7.87/10.
Thesis: *Achievable Rates of Wireless Channels in the Finite Block-Length Regime with a Finite Constellation Input*. Supervisor: Dr. Shahid Mehraj Shah. Grade A.

## Research

2026–present **JT gravity coupled to a bath, and the Page curve**
Two-dimensional (JT) gravity coupled to matter and then to a bath: algebras of observables, generalized entropy, and the Page curve of the radiation.

## Examinations and admissions

2025 **JEST**: All-India rank 170.

2025 **TIFR GS**: qualified; offers of admission from TIFR Hyderabad and IMSc Chennai.

After school **SAT**: admitted to the University of Wisconsin–Madison (BSc Physics with Honours), and to the University of Illinois Chicago and the University of Delaware (Mechanical Engineering).

## Notes

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| 2026 | **Path Integrals, Parts I–III** (171 pages). Typeset and expanded notes on path integrals in quantum mechanics, relativistic quantum mechanics and quantum field theory. faizan-bari.github.io/notes.html |

## Independent study

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| QFT | Peskin and Schroeder, chapters 1–4 and the path-integral chapter. |
| GR | Tensors, topology and differential geometry up to the Einstein equations; David Tong's lectures on general relativity, with problem sets. |
| CFT | Foundations of conformal field theory. |

## Skills

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| Computing | Python for numerical work; LaTeX. |